

CritNeT: A Critical Neurons Toolkit

Abstract

The toolkit is publicly available at <https://github.com/iNLP-Lab/critnet>.

1 Critical Neurons Detection

Drawing on recent interpretability and network pruning methodologies (Wei et al., 2024; Chen et al., 2025; Leng and Xiong, 2025), we quantify the degree of specialization within a model by identifying its critical neurons. Let $\mathcal{W}$ denote the complete parameter set of the model. For a given weight matrix $W \in \mathcal{W}$, let w_i denote the parameter vector associated with an individual neuron i (e.g., a specific *row* or *column* within W).

To evaluate a neuron’s functional importance for a given domain, we approximate the expected change in the loss function $\mathcal{L}$ over a domain-specific dataset $\mathcal{D}$ if w_i were to be deactivated (i.e., pruned to a zero vector, $\mathbf{0}$). Using a first-order Taylor expansion, this impact score $\mathcal{I}_t(w_i)$ can be efficiently estimated as:

$$\begin{aligned} \mathcal{I}_t(w_i) &= |\mathcal{L}(\mathcal{W}, \mathcal{D}) - \mathcal{L}(\mathcal{W} \mid w_i = \mathbf{0}, \mathcal{D})| \\ &\approx \left| w_i^\top \nabla_{w_i} \mathcal{L}(\mathcal{W}, \mathcal{D}) \right|. \end{aligned} \quad (1)$$

2 Utility, Refusal, and Safety Neurons

Language models often entangle general capabilities with alignment guardrails. To disentangle these, we denote $\mathcal{N}_u^p$ and $\mathcal{N}_r^q$ as the top- p and top- q most critical neurons detected using a utility corpus $\mathcal{D}_u$ (representing helpfulness on benign queries) and a refusal corpus $\mathcal{D}_r$ (representing safety on adversarial queries), respectively.

We define *safety neurons*—neurons uniquely specialized for safety-awareness without contributing to general reasoning—as the set difference:

$$\mathcal{N}_s = \mathcal{N}_r^q \setminus \mathcal{N}_u^p. \quad (2)$$

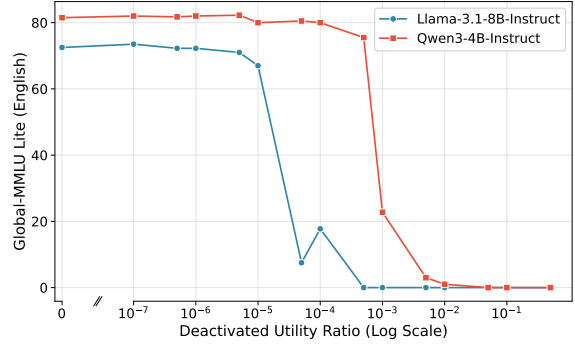


Figure 1: Deactivating a minuscule fraction of utility neurons $\mathcal{N}_u^p$ ($p < 1\%$ of model parameters) completely breaks down the language understanding and reasoning abilities of the models.

2.1 Experimental Setup for Deactivation

Detection Corpora For the utility corpus ($\mathcal{D}_u$), we construct a 1,630-sample English dataset comprising 1,000 LIMA conversational pairs (Zhou et al., 2023) and 630 s1k-1.1 reasoning samples (Muennighoff et al., 2025) distilled from DeepSeek-R1.¹ For the safety corpus ($\mathcal{D}_r$), we employ 1,000 diverse harmful prompt-refusal response pairs selected by Bianchi et al. (2024).

Models and Evaluation We employ two instruction-tuned models, Llama-3.1-8B-Instruct and Qwen3-4B-Instruct. We evaluate their behavioral alignment along two axes: utility and safety. To measure core utility, we utilize the English subsets of four comprehensive benchmarks: ARC-E (Han et al., 2025) and PolyMath (Wang et al., 2025) for reasoning, and Global-MMLU-Lite (Singh et al., 2024) for world knowledge. For safety evaluation, we employ the English subset of the MultiJail benchmark (Deng et al., 2024), computing the Attack Success Rate (ASR) via the open-source HarmBench LLaMA-2-13B classifier (Mazeika et al.,

¹We filter s1k-1.1 to include only samples marked as correctly distilled from DeepSeek-R1.

Table 1: Neuron Deactivation Results (English only). Safety Deactivation (Safety D.) preserves utility while severely degrading safety, whereas Random Deactivation (Random D.) affects the model at a smaller scale.

Method	Utility $\uparrow$			Safety $\downarrow$
	ARC-E	PolyM	MMLU	M-Jail
Llama-3.1-8B-Instruct				
Original	94.3	47.2	73.2	10.2
Random D.	82.2	32.0	58.2	06.4
Safety D.	91.9	24.8	66.2	55.6
Qwen3-4B-Instruct				
Original	98.0	59.2	82.0	03.5
Random D.	97.4	53.2	76.2	01.6
Safety D.	96.8	50.8	74.2	61.6

2024).

2.2 The Fragility of LMs under Utility and Safety Neurons

As shown in Figure 1, deactivating even a minuscule fraction of utility neurons leads to a catastrophic collapse in general reasoning. For both evaluated models, pruning less than 0.1% (a ratio of 10^{-3}) of utility neurons causes performance on Global-MMLU to plummet to near zero. The steep degradation curve highlights that core language understanding is deeply concentrated within these utility-critical parameters, demonstrating the extreme fragility of general capabilities when essential weights are perturbed.

Consequently, isolating pure safety neurons requires an extensive grid search to balance the utility ratio p and refusal ratio q (Figure 2). The heatmaps reveal a distinct boundary: lower values of the refusal ratio combined with higher values of the utility ratio yield a higher attack success rate, indicating effective isolation of safety-specific neurons. We empirically select $p = 0.14, q = 0.04$ for Llama-3.1-8B-Instruct (yielding safety neurons $|\mathcal{N}_s|$ equivalent to 1.8% of the parameter size) and $p = 0.13, q = 0.06$ for Qwen3-4B-Instruct (yielding 1.5%).

Targeted Deactivation Preserves Utility but Breaks Safety. Table 1 demonstrates the precision of our isolated set $\mathcal{N}_s$. Randomly deactivating a comparable 1.5% of neurons moderately impairs general capabilities (e.g., Llama-3.1 MMLU drops from 73.2 to 58.2), but this degradation is mild compared to deactivating utility neurons, and it even decreases harmfulness (ASR drops to 6.4%). Con-

Table 2: Attack Success Rate (lower is better) with Fine-Tuning Attacks on MultiJail-En.

Method	Original	Full Tuning	Frozen-Tuning
Llama-3.1-8B-Instruct	10.2	21.6	24.4
Qwen3-4B-Instruct	03.5	20.0	17.1

versely, deactivating specifically the safety neurons preserves the model’s core utility to a high degree (MMLU remains at 66.2 for Llama-3.1 and 74.2 for Qwen3) while drastically increasing vulnerability. The ASR on MultiJail surges from 10.2% to 55.6% for Llama-3.1, and from 3.5% to 61.6% for Qwen3. This behavioral dichotomy confirms that these specific neurons act as centralized gatekeepers for model alignment.

Fine-Tuning Attacks Bypass Safety Neurons.

Furthermore, we perform fine-tuning on the English corpus of 1,630 benign examples to evaluate alignment resilience.² Table 2 highlights the profound brittleness of these safety mechanisms under adversarial fine-tuning conditions. Standard full fine-tuning significantly compromises alignment, driving the ASR to 21.6% for Llama-3.1 and 20.0% for Qwen3. Even when employing frozen-tuning, which explicitly keeps the detected safety neurons untouched, attackers can easily bypass the model’s intrinsic safety guardrails. This structurally constrained tuning pushes the ASR for Llama-3.1-8B-Instruct from 10.2% up to 24.4%, and for Qwen3-4B-Instruct from 3.5% to 17.1%. These results indicate that preserving safety weights is insufficient; malicious tuning can simply route around these frozen safety pathways to generate harmful outputs.

3 Foundational Neurons and Parameter-Efficient Fine-Tuning

Given a set of languages $\mathcal{T}$, let $\mathcal{N}_t$ denote the set of the top- p most critical neurons for language $t \in \mathcal{T}$, selected based on their impact score $\mathcal{I}_t(w_i)$. By fixing a uniform ratio p across all evaluated languages, we define the **Foundational Neurons** ($\mathcal{N}_{\text{all}}$) as the union of these critical language-specific sub-networks:

$$\mathcal{N}_{\text{all}} = \bigcup_{t \in \mathcal{T}} \mathcal{N}_t. \quad (3)$$

²Fine-tuning details in Appendix A.

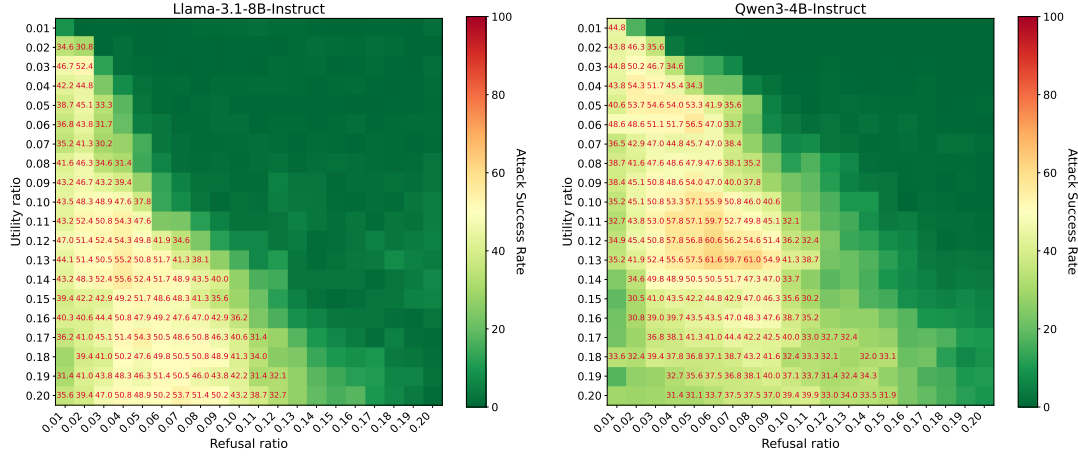


Figure 2: Attack success rate over varying utility (p) and refusal (q) ratios for safety neuron isolation. Lower values of the refusal ratio combined with higher values of the utility ratio yield a higher attack success rate, indicating effective isolation of safety-specific neurons.

3.1 Critical Neurons Tuning as PEFT

Given the global set of foundational neurons $\mathcal{N}_{\text{all}}$, we bypass full fine-tuning and explicit weight matrix partitioning ($W \in \mathbb{R}^{d_{\text{out}} \times d_{\text{in}}}$). Instead, we formulate the optimization as an additive, structurally sparse delta-tuning method. For an input hidden state h , the pre-trained base weight matrix W remains strictly frozen, while we introduce a highly sparse, trainable delta matrix ΔW :

$$h(W + \Delta W)^\top = hW^\top + h\Delta W^\top. \quad (4)$$

To ensure memory and computational efficiency, ΔW is never instantiated as a dense matrix. Let $k = |\mathcal{N}_{\text{all}}|$ denote the number of targeted foundational neurons. We parameterize the trainable subset ΔW_{train} based on the structural orientation of the identified neurons:

- **Output Neurons (Row Mode):** When critical neurons correspond to rows in W , we define $\Delta W_{\text{train}} \in \mathbb{R}^{k \times d_{\text{in}}}$. The linear update is computed dynamically and injected exclusively into the corresponding indices of the output activations: $y_{\mathcal{N}_{\text{all}}} = (hW^\top)_{\mathcal{N}_{\text{all}}} + h\Delta W_{\text{train}}^\top$.
- **Input Neurons (Column Mode):** When critical neurons correspond to columns in W , we define $\Delta W_{\text{train}} \in \mathbb{R}^{d_{\text{out}} \times k}$. The update is computed by slicing the input states at the critical indices prior to projection: $y = hW^\top + h_{\mathcal{N}_{\text{all}}} \Delta W_{\text{train}}^\top$.

In both configurations, the trainable subset ΔW_{train} is initialized to zero, ensuring the initial forward pass perfectly preserves the pre-trained

base model’s representations. Because k constitutes a marginal fraction of the layer dimension ($< 10\%$), this selective routing natively scales as a highly efficient Parameter-Efficient Fine-Tuning (PEFT) strategy.

3.2 Experiments and Analysis

Training Data Preparation We employ the same 1,630-sample English corpus described in Section 2. To construct multilingual tuning data, we utilize Gemini 2.0 Flash to translate this corpus into Chinese, Arabic, and Swahili. The resulting dataset spans four typologically diverse languages (en, zh, ar, sw), effectively covering the high- to low-resource spectrum.

Language Critical Neuron Detection We apply a uniform ratio $p = 1\%$, or $p = 5\%$ across the language set $\mathcal{T} = \{\text{en, zh, ar, sw}\}$. This yields a set of foundation neurons $\mathcal{N}_{\text{all}}$ comprising nearly 3% and 11% of the model’s total parameters. For a rigorous comparison, we baseline our method against LoRA (Hu et al., 2022) using ranks $r = 64$ and $r = 512$.

Evaluation Metrics We extend the utility evaluation suite introduced in Section 2 to a comprehensive multilingual setting. We assess cross-lingual generalization across 13 languages, partitioned into 4 *seen* languages (included in the translation pipeline) and 9 *unseen* languages (bn, de, es, fr, id, it, ja, ko, pt).

Superior Cross-Lingual Transfer and Mitigation of Catastrophic Forgetting. Table 3 details the comparative performance of our founda-

Table 3: Parameter-Efficient Fine-Tuning results. Neuron-based tuning consistently outperforms standard LoRA at equivalent parameter counts (bold indicates superior performance to the corresponding LoRA baseline).

Method	%Param	4 Seen Languages				9 Unseen Languages			
		ARC-E	PolyM	MMLU	Avg.	ARC-E	PolyM	MMLU	Avg.
Llama-3.1-8B									
Base	00.0%	59.3	12.5	44.1	38.6	65.8	13.8	44.7	41.4
Full	100 %	60.9	19.5	44.9	41.8	66.3	20.4	45.6	44.1
LoRA	02.0%	57.0	16.0	42.8	38.6	63.5	18.9	44.2	42.2
Neuron	02.3%	59.6	19.8	44.4	41.3	65.7	24.0	46.0	45.2
LoRA	14.3%	58.1	16.8	43.8	39.6	64.5	19.9	45.1	43.2
Neuron	10.4%	63.2	20.7	46.7	43.5	69.0	23.7	48.5	47.1
Qwen3-4B-Base									
Base	00.0%	75.8	38.1	54.9	56.3	90.7	42.5	61.5	64.9
Full	100 %	75.9	45.0	58.0	59.6	88.4	48.0	62.3	66.2
LoRA	03.2%	74.4	41.4	55.8	57.2	86.4	44.4	61.6	64.1
Neuron	02.3%	75.7	42.6	56.6	58.3	88.9	49.0	62.2	66.7
LoRA	20.8%	74.5	42.0	57.3	57.9	86.6	45.9	60.7	64.4
Neuron	10.8%	75.9	44.2	57.6	59.2	88.7	49.4	62.0	66.7

tional neuron tuning strategy. By strictly confining updates to the foundational neurons ($\mathcal{N}_{\text{all}}$), our method consistently eclipses standard LoRA across multiple parameter budgets. For Llama-3.1-8B, tuning a mere 2.3% of the parameters via foundational neurons yields an average score of 41.3 on seen languages and 45.2 on unseen languages, comfortably surpassing a comparable 2.0% LoRA configuration (38.6 and 42.2, respectively).

Crucially, as we scale the parameter budget to $\sim 10\%$, our approach exhibits remarkable resilience against catastrophic forgetting—a common pitfall of full-parameter fine-tuning. On the 9 unseen languages, the 10.4% neuron-tuned Llama-3.1-8B achieves an average score of 47.1. In contrast, full fine-tuning yields a lower average of 44.1, barely improving upon the zero-shot Base model (41.4). This underscores the structural efficacy of restricting gradient updates to language-critical pathways: it preserves the model’s foundational knowledge while facilitating vastly superior cross-lingual generalization compared to standard dense or low-rank adaptations.

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A Experimental Details

This is an appendix.